

HEALTHCARE DATA CLIENT

Project Overview & Business Problem

Healthcare Data Quality Automation and Exception Reporting

Project Attribute	Confirmed Detail
Engagement Period	Jul-Oct 2026
Role	Data Analyst
Primary Stakeholder	Healthcare Analyst
Processing Scale	Approximately 250,000 records per batch
Observed Exception Rate	2%
Review Turnaround	Reduced from 6 hours to 2.5 hours per file

Document purpose: Defines the business problem, data-quality scope, analytical approach and confirmed outcomes for the Healthcare Data Client engagement.

1. Executive Overview

The engagement focused on improving the review of large healthcare data files through automated data-quality validation and exception reporting. Approximately 250,000 records were processed per batch across healthcare data covering claims, patients, appointments, providers, billing, transactions, clinical records and insurance data.

The solution applied repeatable validation checks, isolated records requiring attention, assigned exception reason codes and generated review outputs. The analysis surfaced a 2% exception rate and reduced review turnaround time from 6 hours to 2.5 hours per file.

2. Business Problem

Large healthcare data files required extensive review before downstream use. The key challenge was to identify data-quality exceptions consistently and reduce the amount of time spent reviewing each file.

- High record volume made complete manual review time-intensive.
- Multiple data-quality conditions needed to be checked consistently across each batch.
- Exceptions needed to be separated from valid records so reviewers could focus on records requiring attention.
- Review output needed enough context, including a reason code, to support investigation.

AS-IS boundary: The exact historical manual workflow is not retained. Any detailed AS-IS process diagram in the portfolio will be presented as a reconstruction rather than as a verbatim client process.

3. Business Objectives

Objective	Business Intent
Automate repeatable validation	Apply consistent data-quality checks across large healthcare files.
Improve exception visibility	Identify and isolate records that fail one or more validation rules.
Support targeted review	Provide exception outputs and reason codes so reviewers can focus on affected records.
Reduce review turnaround	Shorten the time required to review each file.
Provide reporting visibility	Use analytical reporting and Power BI to support data-quality monitoring.

4. Stakeholder & User Need

The primary stakeholder was a Healthcare Analyst. The reporting and validation process needed to support efficient review of large healthcare datasets while keeping exception logic understandable and traceable.

5. Data Landscape

Area	Confirmed Scope
Data domains	Claims, patients, appointments, providers, billing, transactions, clinical records and insurance data
Source formats	Excel, CSV and SQL/database
Batch size	Approximately 250,000 records per batch
Analysis tools	Python, SQL, Excel, Power Query and Power BI

6. Data Quality Scope

Validation Area	Confirmed Check
Duplicate detection	Identify duplicate records.
Completeness	Identify null or missing values.
Date validity	Identify invalid dates.
Format validation	Identify incorrect formats.
Identifier validation	Identify invalid IDs.
Range validation	Identify out-of-range values.
Code consistency	Identify inconsistent codes.
Cross-field validation	Identify mismatches between related fields.

7. Exception Management

Records failing validation were flagged for review rather than being silently corrected or discarded. Exception handling included a separate exception file, table or report, visual highlighting and an assigned reason code.

Exception Rate: Records failing one or more validation rules / Total records processed x 100. The observed exception rate for the engagement was 2%.

8. Automated Validation Flow

Step	Process
1. Ingest	Receive data from the supported file or database source.
2. Validate	Apply the defined data-quality checks.
3. Flag	Identify records failing one or more checks.
4. Classify	Assign an exception reason code and highlight the affected record.
5. Output	Generate the exception file, table or report for review.
6. Review	Healthcare analyst reviews the targeted exception output.

9. Solution Components

Component	Role in Solution
Python	Automation and data-quality validation.
SQL	Data access, querying and validation support.
Excel	File-based review and supporting analysis.
Power Query	Data preparation and transformation for reporting.
Power BI	Data-quality and exception reporting visibility.

10. Confirmed Outcomes

Outcome	Result
Processing scale	Approximately 250,000 records per batch
Exception visibility	2% exception rate surfaced
Review turnaround	Reduced from 6 hours to 2.5 hours per file
Time reduction	3.5 hours less review time per file
Derived turnaround reduction	Approximately 58.3%, calculated from the confirmed 6-hour and 2.5-hour values

Outcome boundary: The 58.3% figure is a derived reduction in review turnaround time. It should not be presented as a productivity increase, labor-cost saving or reduction in healthcare errors without separate evidence.

11. BA / DA Contribution

The Data Analyst role combined healthcare data review, validation-rule implementation, exception analysis, automation, reporting and quality assurance. Requirements, business rules, testing and documentation formed part of the engagement.

Contribution Area	Project Contribution
Requirement interpretation	Translated healthcare analyst review needs into validation and reporting requirements.
Data-quality logic	Implemented repeatable checks for the confirmed exception categories.
Automation	Built a repeatable ingest-to-exception-output validation process.
Exception analysis	Separated affected records and supported reason-based review.
Reporting	Used Power BI alongside analytical tools for data-quality visibility.
Testing and documentation	Supported validation, testing and documentation of the solution.

12. Evidence & Confidentiality Boundaries

Supported Project Evidence	Not Asserted Without Evidence
Approximately 250,000 records per batch	Exact patient, claim, provider or financial values
2% exception rate	Exact exception counts by rule unless retained separately
6 hours to 2.5 hours per file	Labor-cost savings or productivity claims
Confirmed validation categories	Unremembered healthcare-specific adjudication or clinical rules
Excel, CSV and SQL/database sources	Exact historical schema, table names or field definitions
Python, SQL, Excel, Power Query and Power BI	Exact historical code where not retained
Exception outputs with reason codes	Fabricated PHI, patient identifiers or confidential client data

13. Project Narrative

Interview-ready summary: I worked as a Data Analyst on a healthcare data-quality engagement processing approximately 250,000 records per batch. The objective was to make large-file review faster and more consistent. I used Python, SQL, Excel, Power Query and Power BI to apply automated validation checks for duplicates, nulls, invalid dates, incorrect formats, invalid IDs, out-of-range values, inconsistent codes and cross-field mismatches. Records failing validation were isolated, highlighted and assigned reason codes for targeted review. The solution surfaced a 2% exception rate and reduced review turnaround from 6 hours to 2.5 hours per file.